Growth Regimes in the United States since 1920: Structural Breaks, Fiscal Structure, and the Productivity–Pay Divergence

Diogo Ribeiro*

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Abstract

We characterise the annual growth process of United States real GDP from 1920 onward as a sequence of statistically identified regimes rather than a single trend. Using a Maddison-based long-run real GDP series validated against BEA national accounts, we first show that log GDP is empirically indistinguishable from a unit-root process while its growth rate is stationary, so growth—not the level—is the appropriate object for regime analysis. A dynamic-programming segmentation with information-criterion selection identifies 5 global regimes; because a single pooled variance lets the extreme 1929–1945 episode dominate, we apply a recursive refinement that re-segments each regime on its own scale, yielding 6 regimes and resolving the long postwar era into a high-growth 1950–2000 phase and the post-2001 slowdown. Parametric-bootstrap break tests, break-date confidence intervals, and a robustness grid over tuning choices establish which breaks are firm and which are fragile. We then situate the regimes against federal fiscal structure, the dynamic response of growth to narrative tax-regime shocks, and the widening gap between real GDP per capita and real median earnings. The exercise is deliberately descriptive: it treats statistical breaks as structure to be interpreted, not as identified causal effects.

Keywords: United States GDP, growth regimes, structural breaks, unit roots, local projections, fiscal policy, real wages, reproducible research

JEL classification: C22, E32, N12, O47, E62.

Reproducible from the repository pipeline; raw downloaded data are excluded from version control. Every statistic in the text is a machine-generated macro derived from the model outputs in `data/models/`.

1 Introduction

The long-run growth of the United States economy is routinely summarised by a single average rate. That summary is analytically convenient but substantively misleading: a century that contains the

*All estimates, tables, and figures are reproducible from the accompanying open-source pipeline (`us-gdp-regime-1920`); every quantitative statement in the text is generated programmatically from the model outputs. Correspondence: `hansolo.dj@gmail.com`.

Great Depression, wartime mobilisation, the postwar boom, and the post-2000 slowdown is not well described by one number. This paper asks a deliberately simple question—did the United States grow at one roughly constant speed after 1920, or is its growth history better read as a sequence of distinct regimes?—and answers it with a transparent, fully reproducible empirical workflow.

Our approach has four components. First, we establish the correct object of study. A linear trend fits log real GDP almost perfectly in sample ($R^2 = 0.987$), but that fit is a well-known artefact of persistence: augmented Dickey–Fuller [12] and KPSS [8] tests together indicate that the log level is consistent with a unit root while the growth rate is stationary. Regimes are therefore estimated on growth. Second, we detect regimes with an exact dynamic-programming segmentation of the growth series and select their number by information criterion. Because a single global variance lets the violent 1929–1945 years dominate, we introduce a recursive refinement that re-segments each regime on its own variance scale; this is what separates the postwar boom from the post-2001 slowdown that a pooled model conceals. Third, we quantify the uncertainty of the regimes with sequential structural-break tests in the spirit of Bai and Perron [2, 3], bootstrap confidence intervals for the break dates, and a robustness grid across tuning choices. Fourth, we relate the regimes to federal fiscal structure, to the dynamic response of growth to narrative tax-regime shocks estimated by local projections [7], and to the divergence between aggregate output per capita and the earnings of the median worker.

Contributions. The paper makes three modest contributions. (i) It documents that the standard deterministic-trend summary of U.S. growth is statistically inappropriate, and provides the unit-root and stationarity evidence that motivates a growth-regime treatment. (ii) It shows that a naive global segmentation understates postwar heterogeneity because of variance pooling, and that a simple recursive refinement—corroborated by three independent criteria and by a subsample break test—recovers the post-2000 slowdown as a distinct regime. (iii) It integrates fiscal, tax-dynamic, and distributional evidence into the same reproducible framework while maintaining a strict separation between descriptive timing, lagged association, and causal identification.

Related literature. The empirical strategy draws on the multiple structural-change framework of Bai and Perron [2, 3] and on the observation of Perron [10] that unmodelled breaks bias unit-root tests toward non-rejection—a caveat we take seriously in interpreting the level diagnostics. Dynamic responses are estimated by local projections [7] with Newey–West covariance [9]; the distinction between plausibly exogenous and endogenous tax changes follows the narrative-identification tradition of Romer and Romer [11]. Substantively, the post-2000 regime connects to the debate over the productivity slowdown [6, 5] and secular stagnation [14], while the output–earnings divergence connects to work on the productivity–pay gap [13] and rising earnings inequality [1]. The long-run series is the Maddison Project Database [4].

Roadmap. Section 2 describes the data and validation. Section 3 sets out the empirical strategy. Section 4 presents the trend diagnostics, regimes, break inference, and postwar decomposition; Section 5 reports robustness. Sections 6 and 7 add the fiscal, tax-dynamic, and distributional layers. Section 8 discusses the findings, Section 9 the limitations, and Section 10 concludes.

2 Data

2.1 Core GDP series and validation

The primary series is annual United States real GDP from 1920 onward, constructed from the Maddison Project Database 2023 [4] as real GDP per capita multiplied by population. Maddison is designed for long-run historical comparison and is the natural source for a 1920 starting point; the growth-rate analysis is invariant to the constant unit rescaling implied by the population units. The modern benchmark is the Bureau of Economic Analysis annual real GDP series (FRED `GDPCA`), available from 1929.

Over the 1930–2022 overlap (93 annual observations), Maddison-derived and BEA growth rates correlate at 0.955, with a mean absolute difference of 0.63 and a root mean squared difference of 1.47 percentage points. The largest discrepancies (Appendix A) fall in the wartime years, when historical and national-accounts output concepts diverge most. The agreement is close enough for regime-level analysis while reminding the reader that the two sources are distinct objects.

2.2 Fiscal, tax, and distributional data

Three further layers provide context. The fiscal layer uses FRED/OMB federal debt, receipts, outlays, deficit, and interest-outlay ratios as a share of GDP. The tax layer uses a curated catalog of broad federal tax-law changes with signed ordinal shock values and narrative classifications distinguishing plausibly exogenous long-run reforms from deficit-driven, wartime, and countercyclical changes. The distributional layer adds BEA real GDP per capita, BLS real median weekly earnings and real hourly compensation, the composition of federal receipts, statutory top and bottom income-tax rates, and BLS Consumer Expenditure federal income-tax rates by income quintile. These series answer different questions and are not interchangeable welfare measures.

3 Empirical Strategy

3.1 Trend and unit-root diagnostics

The long-run trend is the ordinary least squares regression of log real GDP on a linear time trend,

$$\log(GDP_t) = \alpha + \beta\, t + \epsilon_t, \tag{1}$$

with the annualised trend rate $e^{\beta} - 1$. Because annual macroeconomic residuals are serially correlated, β is reported with Newey–West heteroskedasticity- and autocorrelation-consistent standard errors [9]. A high R^2 in (1) is not evidence for a deterministic trend: a near-integrated series is mechanically well fit by a line. We therefore test the log level and the growth rate with the augmented Dickey–Fuller test [12], whose null is a unit root, and the KPSS test [8], whose null is stationarity. We read the two jointly, and we note the caution of Perron [10] that unmodelled structural breaks bias ADF toward failing to reject a unit root.

3.2 Growth regimes and recursive refinement

Regimes are a piecewise-constant mean model for annual growth,

$$g_t = \mu_j + u_t, \qquad t \in \text{regime } j, \tag{2}$$

where the segmentation into contiguous regimes is chosen to minimise a Gaussian information criterion. For a partition into k segments with within-segment sum of squared residuals SSR_k and n observations, the criterion is

$$\mathrm{IC}(k) = n \log\left(\frac{\mathrm{SSR}_k}{n}\right) + p_k\, c(n), \qquad p_k = 2k, \tag{3}$$

with $c(n) = \log n$ for BIC and $c(n) = 2$ for AIC; the parameter count $p_k = 2k$ comprises k segment means, $k-1$ break locations, and one residual variance. The globally optimal partition for each k is obtained exactly by dynamic programming over prefix sums, and k is chosen to minimise (3). Each segment is labelled "above" or "below" the full-sample mean; the labels are descriptive.

A single global segmentation imposes one residual variance on the whole century. This is a poor assumption for U.S. growth: the 1929–1945 swings inflate the pooled variance so much that calmer postwar mean shifts never clear the penalty. We therefore apply a *recursive refinement*. Starting from the global fit, each segment is re-segmented on its own observations—and thus its own residual variance—and split whenever (3) supports it, up to a bounded depth. This is the headline model; the global fit is retained as a reference.

3.3 Break inference

The number of regimes is tested, not assumed. For each step from l to $l+1$ segments we form a sequential $\mathrm{supF}(l+1 \mid l)$ statistic from the optimal SSR of the two nested models. Because the break date is estimated, the statistic does not follow a standard F distribution; we obtain its p-value by a parametric bootstrap (199 replications) that simulates Gaussian data from the fitted smaller model [2]. Uncertainty in the break *locations* is quantified separately by a residual bootstrap (499 replications) that re-segments each resample at a fixed number of breaks and reports percentile intervals for each break year.

3.4 Tax-regime dynamics and distributional indices

Dynamic responses of growth to tax-regime shocks are estimated by local projections [7],

$$g_{t+h} = \alpha_h + \beta_h\, s_t + \gamma_h\, g_{t-1} + \gamma_h'\, g_{t-2} + e_{t+h}, \qquad h = 0, \ldots, 10, \tag{4}$$

where s_t is a signed tax-regime shock. Because the horizon-h residual is an $\mathrm{MA}(h)$ process by construction, the Newey–West lag length grows with h. We estimate (4) on all catalogued events and, separately, on the subset classified as plausibly exogenous [11]. Distributional series are indexed to a common base year, $\mathrm{index}_{x,t} = 100\, x_t / x_{base}$, and the headline gap is $\mathrm{index}_{GDPpc,t} - \mathrm{index}_{earnings,t}$.

4 Results

4.1 Trend and unit-root diagnostics

Table 1 reports the log-trend regression. The slope is precisely estimated even under HAC covariance ($t = 36.3$), implying annualised trend growth of 3.16%. As anticipated, this precision reflects persistence rather than a genuine deterministic trend. Table 2 makes the point formal: for the log level, ADF does not reject a unit root (-0.11, $p = 0.993$) and KPSS rejects trend-stationarity ($p = 0.037$); for growth, the verdicts reverse (ADF $p < 0.001$; KPSS $p = 0.100$). Growth is the stationary object, and the appropriate one for regime analysis.

Table 1. Log-trend regression of real GDP, 1920–present

Log-trend slope $\hat{\beta}$ (per year)	0.0311
Newey–West standard error	(0.0009)
HAC t-statistic	36.3
Implied annual growth $e^{\hat{\beta}} - 1$ (%)	3.16
R^2	0.987
Observations	103
Newey–West lags	4

Notes. OLS of log real GDP on a linear time trend, Eq. (1). Newey–West standard error in parentheses. The high R^2 is an artefact of persistence, not evidence of a deterministic trend.

Table 2. Unit-root and stationarity tests

Series	Spec.	ADF stat	ADF p	KPSS stat	KPSS p
Log real GDP level	ct	-0.11	0.993	0.162	0.037
Real GDP growth	c	-6.02	¡0.001	0.071	0.100

Notes. ADF null: unit root; KPSS null: stationarity. Specification `ct` includes a constant and trend; `c` a constant only. Read jointly, the level is difference-stationary and growth is stationary.

Figure 1 plots log real GDP with the fitted trend of Eq. (1). The line tracks the level closely, but the diagnostics above show that this apparent fit is the signature of a near-integrated series rather than evidence of a stable deterministic path.

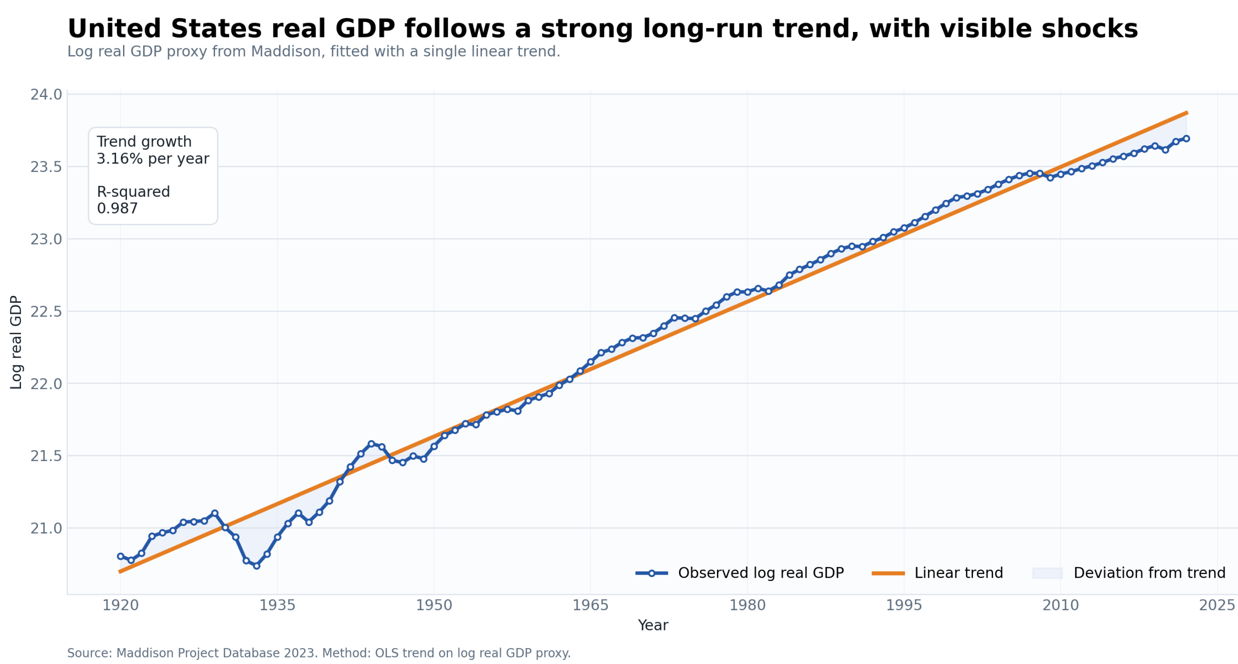

Figure 1. Log real GDP and fitted linear trend. The close fit reflects persistence, not a deterministic trend (cf. Tables 1 and 2). Source: Maddison Project Database 2023.

4.2 Growth regimes

Table 3 reports the 6 regimes of the headline model, plotted in Figure 2. The 1929–1933 contraction ($\approx -5.8\%$) and the 1934–1944 rebound and mobilisation ($\approx +8.1\%$) dominate the early sample. A plain global fit leaves the entire post-1950 era as a single block (5 regimes); the recursive refinement splits it into a high-growth 1950–2000 regime averaging 3.63% and a slower 2001–2022 regime averaging 1.89%, against a full-sample mean of 2.97%.

Table 3. Estimated annual growth regimes (headline recursive model)

Start	End	Years	Mean growth	Label
1921	1928	8	3.19%	Above mean
1929	1933	5	-5.78%	Below mean
1934	1944	11	8.10%	Above mean
1945	1949	5	-1.98%	Below mean
1950	2000	51	3.63%	Above mean
2001	2022	22	1.89%	Below mean

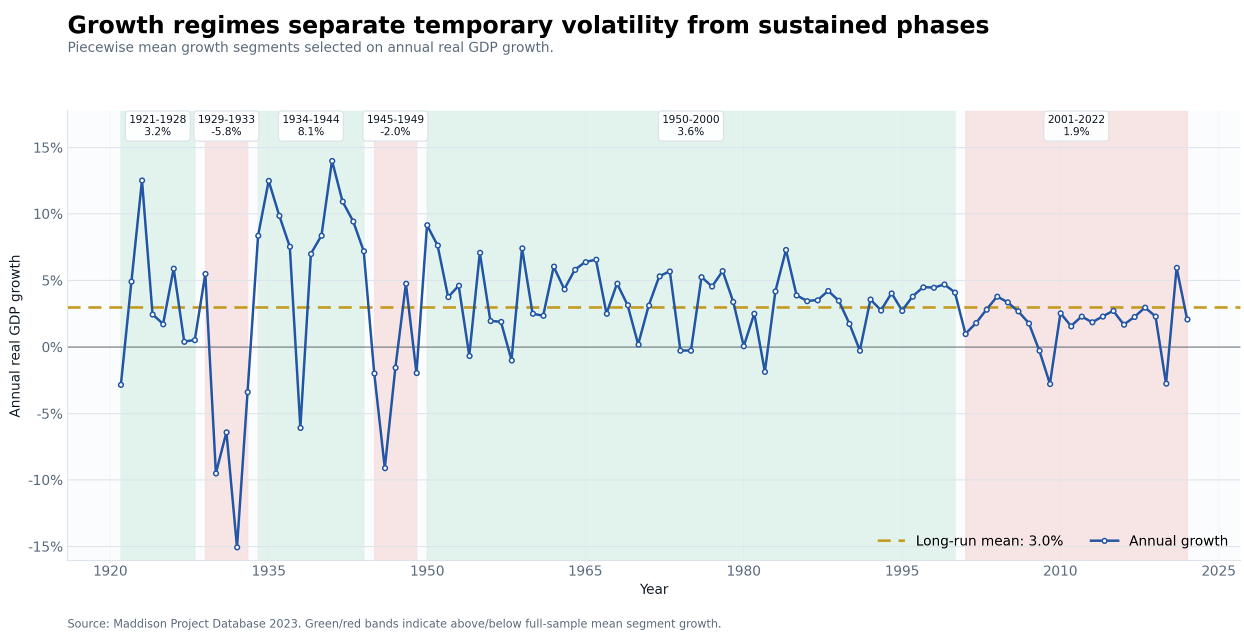

Figure 2. Annual real GDP growth with piecewise-constant regimes. Bands are statistical segments, not claims that any single event caused an entire segment. Source: Maddison Project Database 2023.

4.3 Break significance and dating

Table 4 reports the sequential break tests for the global model. Additional regimes earn small bootstrap p-values up to 5 segments and are insignificant thereafter. This is the formal counterpart of the global information-criterion choice, and it explains why a single-variance model stops before splitting the postwar era: relative to the enormous 1929–1945 residual variance, the postwar mean shift does not clear the global bar. That is a property of the pooled-variance assumption, not evidence that the postwar era is homogeneous.

Table 4. Sequential supF($l+1 \mid l$) break tests

Segments (null)	Segments (alt.)	F statistic	Bootstrap p
1	2	8.43	0.070
2	3	17.04	0.005
3	4	18.08	0.005
4	5	9.58	0.045
5	6	3.74	0.510
6	7	2.64	0.760
7	8	1.48	0.930
8	9	1.05	0.985

Notes. Parametric-bootstrap p-values, 199 replications, simulating from the fitted smaller model.

The break *dates* are unevenly identified (Table 5). The Depression and mobilisation breaks are sharp, whereas the start of the long postwar regime near 1950 has a percentile interval spanning 1950–1988. That imprecision is itself a symptom: the global model is forcing a heterogeneous postwar era through one break and one mean.

Table 5. Bootstrap confidence intervals for the break years (global model)

Break	Point year	90% interval	Std. (years)
1	1929	[1927, 1929]	0.9
2	1934	[1934, 1934]	1.2
3	1945	[1944, 1946]	3.9
4	1950	[1950, 1988]	13.6

4.4 Postwar decomposition

Re-segmenting the postwar era on its own variance scale resolves the ambiguity. Table 6 shows that three independent choices—the postwar subsample under BIC, the postwar subsample under AIC, and the full sample under AIC—all split the era at 2001 into the same 1950–2000 and 2001–2022 regimes. On the postwar subsample the single break at 2001 is statistically significant ($p = 0.045$) while no further postwar break is. This is the post-2000 growth slowdown, and it is the regime that recursive refinement adds to Table 3.

Table 6. Postwar decomposition: three criteria agree on the 2001 split

Method	Early regime	Mean	Late regime	Mean
Postwar subsample, BIC	1950–2000	3.63%	2001–2022	1.89%
Postwar subsample, AIC	1950–2000	3.63%	2001–2022	1.89%
Full sample, AIC	1950–2000	3.63%	2001–2022	1.89%

5 Robustness

Table 7 re-estimates the break years across 7 scenarios: minimum segment sizes of 4, 5, 7, and 10 years; the BIC and AIC criteria; and exclusion of the 1941–1945 wartime window. Break years are clustered within a two-year tolerance and reported with the number of scenarios in which they appear. The Depression trough (1933) recurs in all 7 scenarios, and the late-1920s, mid-1940s, and late-1940s breaks recur in the large majority. The post-2000 slowdown appears only under the AIC scenario in this global grid, consistent with the finding above that a pooled-variance model under-selects it; its support comes instead from the subsample decomposition of Table 6. The message is twofold: the pre-war regime structure is highly robust to tuning, while the postwar split is real but criterion-dependent and is best identified on its own scale.

Table 7. Recurring break years across robustness scenarios

Representative year	Range	Scenarios
1928	1928–1929	5/7
1933	1933	7/7
1944	1944	6/7
1949	1949	4/7
2000	2000	1/7

Notes. Scenarios vary the minimum segment size (4, 5, 7, 10), the criterion (BIC, AIC), and exclude 1941–1945. Break years within two years are clustered.

6 Fiscal Structure and Tax-Regime Dynamics

6.1 Fiscal context

Federal fiscal ratios move sharply across wars, recessions, and crises, but their same-year comovement with growth is modest: gross debt-to-GDP correlates with growth at -0.32 contemporaneously and -0.26 at a one-year lag, with smaller correlations for receipts, outlays, and deficits (Figure 3). These are timing devices, not identified fiscal effects: debt, deficits, and receipts are endogenous to the same shocks that move output.

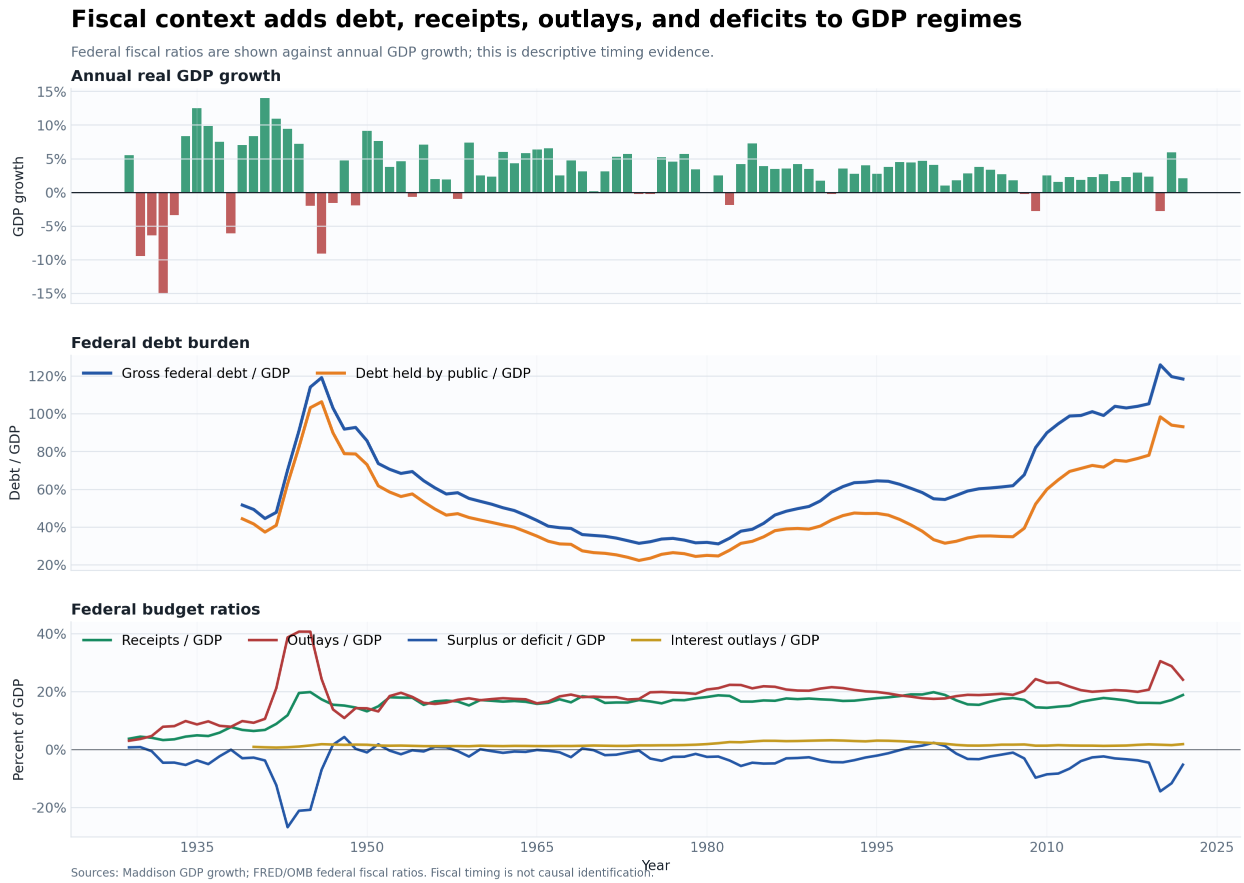

Figure 3. Federal debt, budget ratios, and annual GDP growth. A timing and context device; it does not identify fiscal causality.

6.2 Dynamic response to tax-regime shocks

Table 8 and Figure 4 report local-projection estimates of growth following signed tax-regime shocks, for all catalogued events and for the plausibly exogenous subset [11]. The coefficients are small, switch sign across horizons, and are almost never individually significant; confidence intervals are wide, especially for the exogenous subset, which contains few events. The results should be read as a dynamic screening exercise, not as tax multipliers. Their value is methodological: by allowing delayed responses, they show that the data do not support a simple same-year tax story, consistent with anticipation, phase-ins, and heterogeneous channels.

Table 8. Local projections of GDP growth on tax-regime shocks

	All events		Exogenous subset	
Horizon h	Coef.	(SE)	Coef.	(SE)
0	-1.46	(1.63)	0.50	(1.06)
1	0.01	(1.25)	0.84	(1.32)
2	0.46	(0.85)	-1.73*	(1.00)
3	0.65	(1.03)	0.07	(1.67)
4	-0.72	(1.51)	-1.04	(0.82)
5	0.07	(0.78)	-0.31	(0.91)
6	1.02	(1.52)	3.23	(2.21)
7	0.49	(1.09)	1.92	(1.91)
8	2.38	(1.61)	3.72	(3.20)
9	2.41**	(1.14)	0.89	(1.23)
10	1.50**	(0.59)	1.37	(1.44)

Notes. Coefficient β_h from Eq. (4); Newey–West standard errors with lag growing in h. *, **, *** denote significance at 10%, 5%, and 1%.

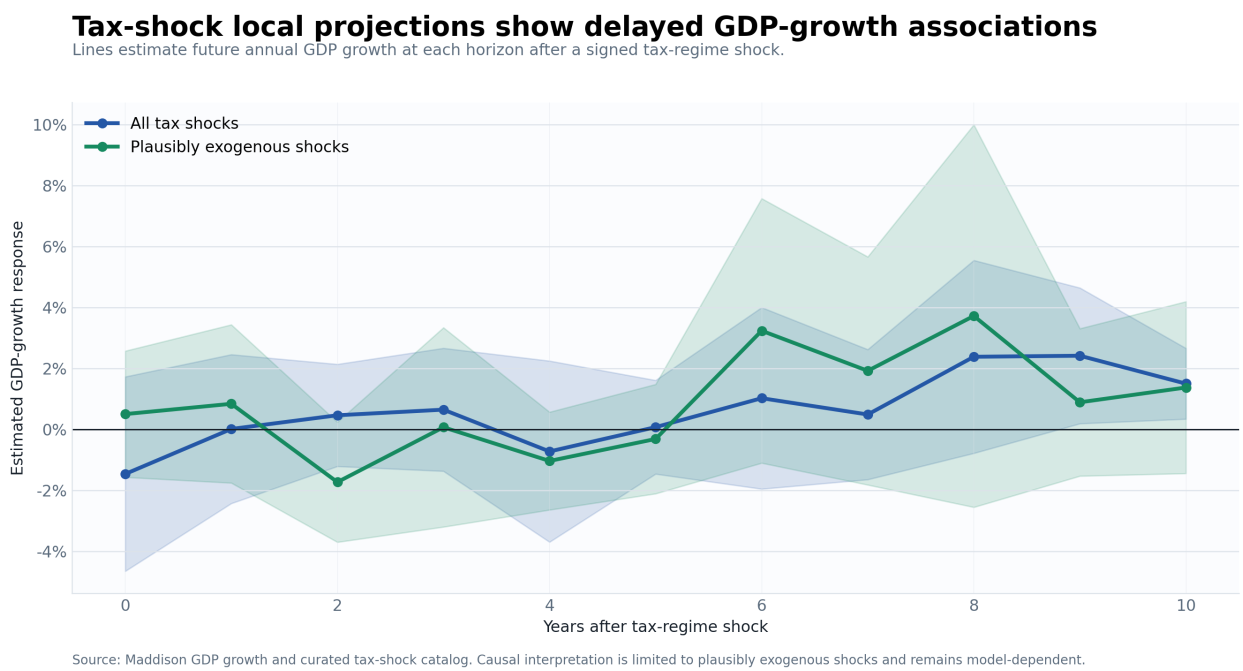

Figure 4. Local-projection estimates after signed tax-regime shocks. The restricted line uses only plausibly exogenous long-run reforms.

7 The Productivity–Pay Divergence

The regime that closes the sample—the post-2000 slowdown—coincides with a long-running divergence between aggregate output and worker earnings. Figure 5 indexes real GDP per capita, real median weekly earnings, and real hourly compensation to 1979. By 2026 real GDP per capita reached 218.4 (more than double its base), while real median weekly earnings reached only 113.4; real hourly compensation (156.8) lies between. Aggregate output per person can rise substantially while the median worker's measured real earnings grow slowly.

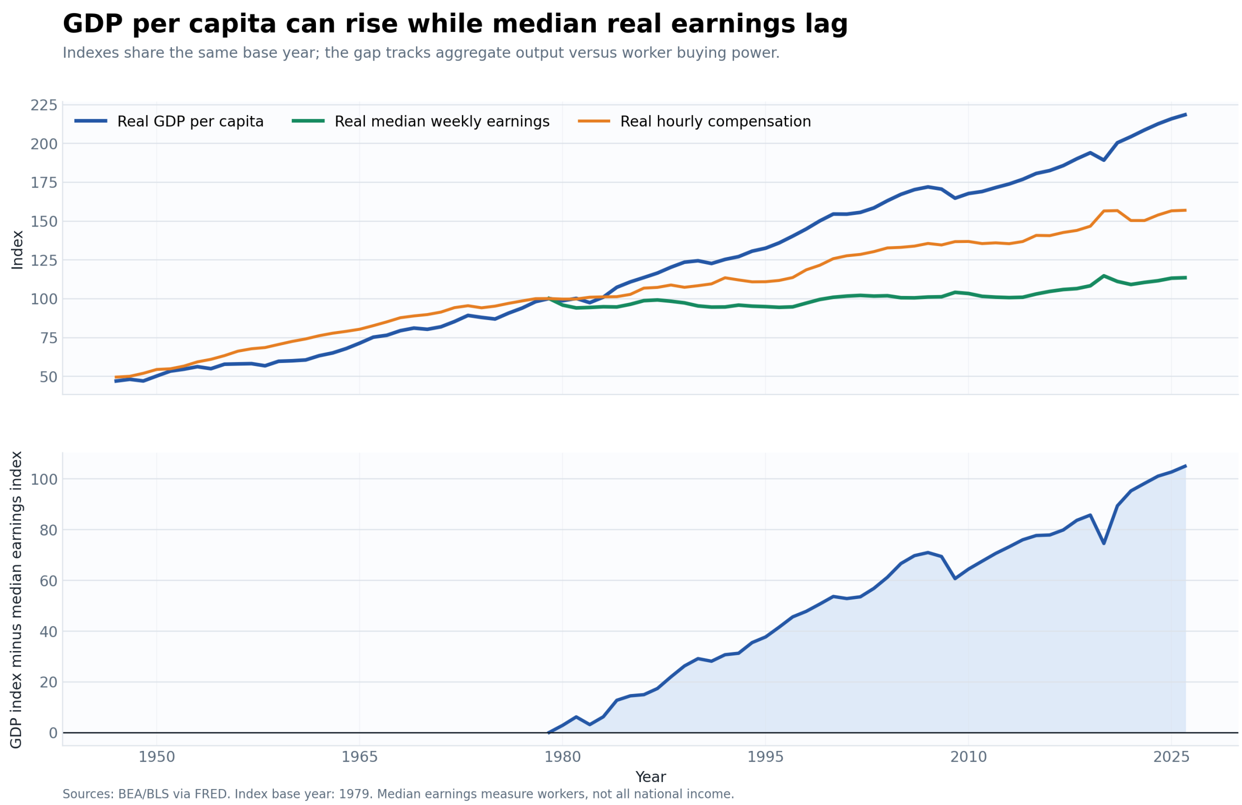

Figure 5. Real GDP per capita, real median weekly earnings, and real hourly compensation, indexed to 1979. The widening gap shows why GDP growth is not equivalent to broad worker buying power.

The composition of the tax system shifted over the same period (Figure 6). Social-insurance contributions rose from 15.1% of the selected federal receipt base in 1947 to 35.1% in 2026, while corporate income taxes fell from 29.6% to 10.1%. Federal income-tax progressivity by quintile, however, did not collapse: the top quintile's average federal income-tax rate exceeded the bottom-four-quintile average by 5.25 points in 1984 and 14.10 points in 2023. These proxies are not a full incidence model; they indicate a shift toward social-insurance financing rather than a simple reduction in income-tax progressivity.

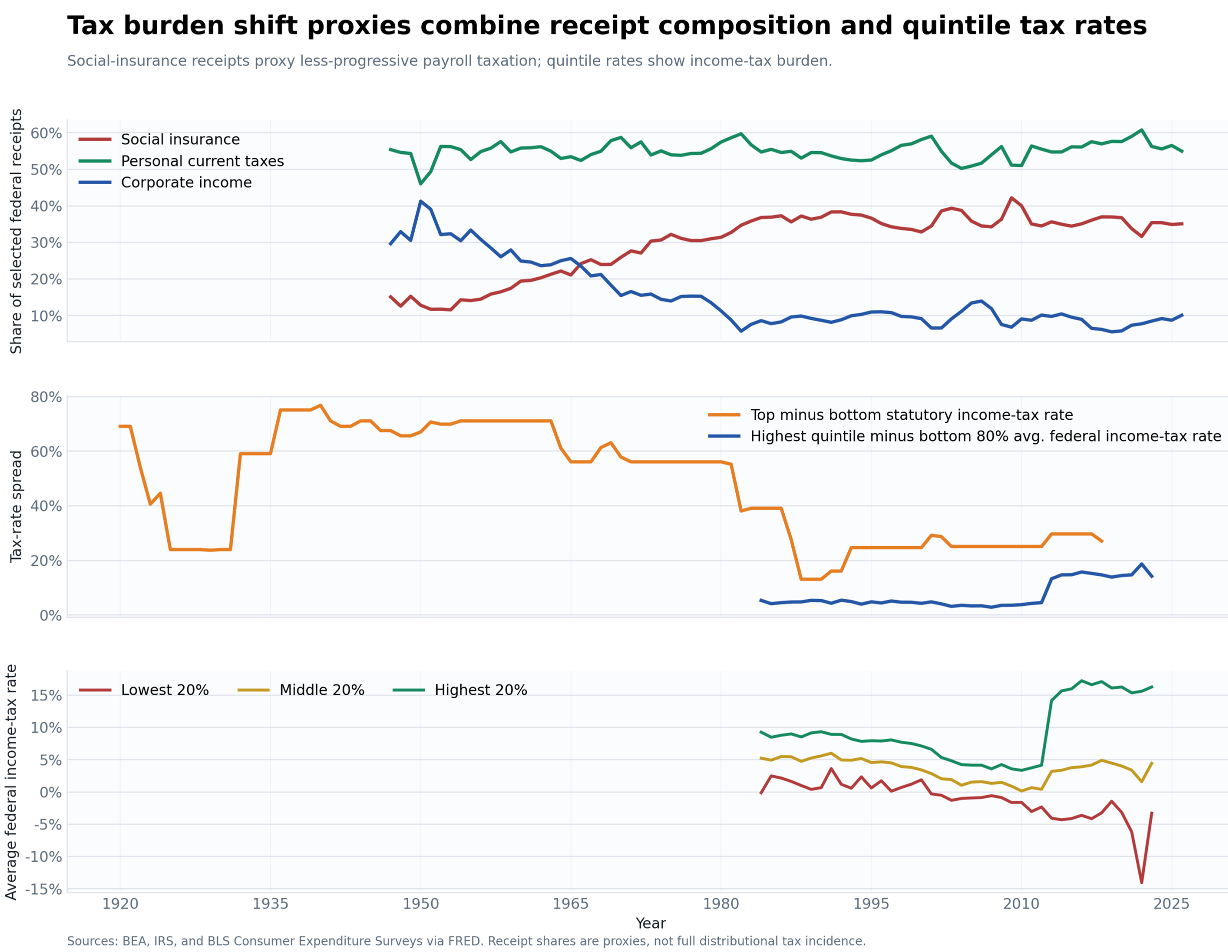

Figure 6. Tax-burden shift proxies: receipt composition and, where available, average federal income-tax rates by income quintile.

8 Discussion

Three findings deserve emphasis. First, the appropriate summary of U.S. growth is a stationary process punctuated by regimes, not a deterministic trend; the unit-root evidence is not a technicality but the reason a regime treatment is warranted, and it echoes the caution of Perron [10] that breaks and unit roots are easily confounded. Second, the modelling choice of one global variance versus regime-specific variances is consequential: it is the difference between a homogeneous 73-year postwar block and a postwar boom followed by a distinct post-2000 slowdown. That slowdown is precisely the phenomenon at the centre of the productivity-slowdown [6, 5] and secular-stagnation [14] debates, and locating it as a data-driven break rather than an assumed date is a useful discipline. Third, the divergence between output per capita and median earnings shows that aggregate growth is an incomplete welfare statistic; the pattern is consistent with the broader productivity–pay [13] and inequality [1] literatures, though our proxies cannot decompose its sources. Throughout, we resist causal readings: the fiscal and tax layers are endogenous to the same shocks that drive output, and the local-projection estimates are screening devices, not multipliers.

9 Limitations

Three limitations bound the interpretation. *Identification*: breakpoints, fiscal ratios, tax laws, and wages all move with macroeconomic conditions; the narrative classification of tax shocks mitigates but does not solve this, and the event sample is small. *Measurement*: Maddison is not identical to BEA national accounts, median weekly earnings are not household income or total compensation, and the tax-burden proxies are not a full incidence model. *Coverage*: the Maddison-based regime sample ends in 2022, while several FRED series in the fiscal and distributional layers extend to 2026 and may include a partial final year; coverage is therefore reported by series rather than assumed uniform.

10 Conclusion

The United States did not grow at one constant speed after 1920. A deterministic trend fits the log level in sample but is statistically inappropriate, because the level is indistinguishable from a unit-root process while growth is stationary. Read through growth, the century resolves into a small number of statistically supported regimes: a 1920s expansion, the Depression, wartime mobilisation, postwar demobilisation, the postwar boom, and the post-2001 slowdown. Recovering that last regime requires taking regime-specific variances seriously; a pooled-variance model conceals it. Placed next to fiscal structure, tax-regime dynamics, and the widening output–earnings gap, the regimes make a simple point precisely: aggregate GDP growth is informative but incomplete, and is best read as a trend-plus-regimes process reported alongside the fiscal and distributional facts it omits.

Reproducibility Statement

The paper is generated from the repository pipeline:

- `us-gdp-regimes run --config config/default.yaml`
- `us-gdp-regimes make-fiscal-context --config config/default.yaml`
- `us-gdp-regimes make-tax-effects --config config/default.yaml`
- `us-gdp-regimes make-distributional-analysis --config config/default.yaml`
- `us-gdp-regimes export-report-numbers --config config/default.yaml`

The final command regenerates `generated_numbers.tex` and the generated tables that this paper inputs, so every quoted statistic is reconciled with the model outputs. Raw downloaded data, model CSVs, figures, and generated tables are reproducible artefacts and are not committed as source.

A Source Validation Detail

Table 9 lists the years with the largest absolute difference between Maddison-derived and BEA (`GDPCA`) annual growth. The largest gaps fall in the World War II years, where wartime output measurement and historical reconstruction diverge most; excluding these years leaves close agreement across the overlap.

Table 9. Largest Maddison–BEA annual growth differences

Year	Maddison (%)	BEA (%)	Difference (pp)
1942	10.91	18.89	-7.97
1943	9.44	17.02	-7.58
1941	13.99	17.71	-3.72
1935	12.47	8.90	3.57
1936	9.87	12.88	-3.02
1938	-6.08	-3.31	-2.77
1946	-9.11	-11.61	2.50
1934	8.37	10.81	-2.44
1937	7.53	5.12	2.41
1932	-15.07	-12.90	-2.17

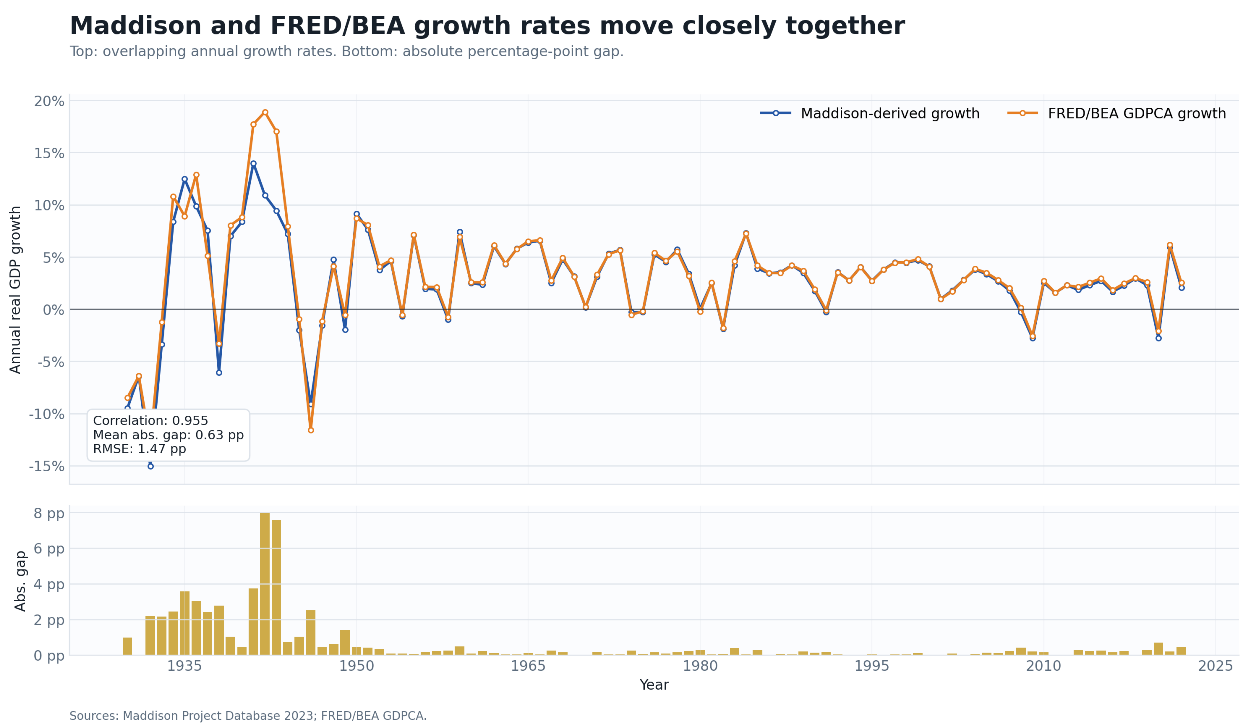

Figure 7. Overlapping annual real GDP growth from Maddison-derived GDP and BEA `GDPCA`. A validation diagnostic, not a causal test.

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[FRED GDPCA] Federal Reserve Bank of St. Louis. Real Gross Domestic Product, annual, `GDPCA`. https://fred.stlouisfed.org/series/GDPCA.

[U.S. OMB] Office of Management and Budget. Historical Tables. https://www.whitehouse.gov/omb/budget/historical-tables/.

[BLS Consumer Expenditure Surveys] U.S. Bureau of Labor Statistics. Consumer Expenditure Surveys. https://www.bls.gov/cex/.

[BEA National Accounts] U.S. Bureau of Economic Analysis. National Income and Product Accounts. https://www.bea.gov/data/gdp/gross-domestic-product.